Product & Architecture Blueprint: A Next-Generation AI SEO Platform

Working name used throughout: "Visio" (placeholder — rename freely) Version 1.0 — July 2026

This blueprint converts the research findings into a buildable product plan: positioning, module map, system architecture, data layering, AI components, frontend/UX system, tech stack, pricing, feasibility, and a phased roadmap with team estimates.

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1. Product Vision & Positioning

One-line vision: The single platform where any business — from a Delhi kirana store to a Fortune 500 — sees its complete visibility across Google, AI answer engines, and agentic commerce, and where the platform doesn't just diagnose problems, it fixes them.

The three bets the product is built on:

1. **Visibility is fragmenting, budgets are not.** Businesses will not buy five tools (rank tracker + technical auditor + content optimizer + GEO tracker + local SEO). The winner unifies SEO + GEO/AEO around one visibility model at one price.
2. **Measurement without execution is theater.** Every incumbent and every GEO startup stops at the dashboard. The defensible moat is an agentic execution layer that pushes approved fixes into the customer's CMS, code, and Google Business Profile.
3. **The next billion SEO customers are not English-first.** Vernacular NLP (starting with Indian languages), region-tiered pricing, and multi-engine coverage (Baidu, Naver, Yandex, Bing) open markets incumbents structurally ignore.

Positioning statement: "The AI visibility platform that fixes what it finds." Against Ahrefs/Semrush: unified AI-era visibility + execution at a fraction of the cost. Against Profound/Peec: everything they measure, plus classic SEO, plus the fix-it layer. Against enterprise suites: 48-hour time-to-value instead of a 3-month ramp.

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2. Product Pillars & Module Map

Pillar A — Unified Visibility Engine (the "see everything" layer)

Module	Sub-modules	Phase
Rank Tracking	Multi-geo/device/language tracking, SERP feature tracking, volatility index, historical SERP snapshots	MVP

Module	Sub-modules	Phase
AI Visibility (GEO)	Prompt-level monitoring across ChatGPT, Perplexity, Gemini, Copilot, AI Overviews, AI Mode; citation source tracking; sentiment & accuracy scoring; Share of Model	MVP
Unified Share of Voice	Single blended score: organic SoV + AI SoV + local SoV, weighted by the customer's traffic mix	MVP
Keyword & Prompt Research	Volume, difficulty, intent classification, semantic clustering, question mining, query fan-out discovery, vernacular/transliterated keywords	MVP
Competitor Intelligence	Keyword gap, citation gap, content gap, entity gap, backlink gap	v1.5
Backlink & Mention Index	Licensed link data initially; unlinked brand-mention tracking (critical for AI citations — mentions correlate ~3x stronger than backlinks with AI Overview visibility)	v1.5

Pillar B — Diagnosis Engine (the "what's wrong" layer)

Module	Sub-modules	Phase
Technical Audit	Cloud crawler (JS rendering), Core Web Vitals, indexability, structured data validation, redirect/canonical analysis, AI-crawler access audit (GPTBot/ClaudeBot/PerplexityBot allow-list analysis)	MVP
Content & Extractability Scoring	Entity/topic coverage, answer-first structure scoring, passage self-containment, E-E-A-T signal detection, content decay alerts	v1.5
Entity & Knowledge Graph Audit	Wikidata/sameAs consistency, schema entity mapping, Knowledge Panel status, cross-web entity corroboration score	v1.5
Log File Analysis	Crawl budget, AI-bot traffic analysis, orphan page detection	v2 (enterprise)
Local SEO Audit	GBP completeness, citation consistency (NAP), review velocity & sentiment	v1.5

Pillar C — Agentic Execution Layer (the moat)

This is the headline differentiator. Every finding in Pillar B maps to an executable action, presented in a **"Fix Queue"** with human-in-the-loop approval:

- **Schema injection:** generate + deploy JSON-LD across thousands of pages via CMS plugin or edge worker (Cloudflare Worker) — no dev ticket needed.
- **Content actions:** AI-drafted rewrites for extractability (answer-first paragraphs, FAQ blocks), meta title/description regeneration at scale, internal-link insertion suggestions applied on approval.
- **Technical fixes:** redirect rules, canonical corrections, hreflang generation, robots.txt AI-crawler policies — deployable via CMS integrations or exportable as PRs (GitHub integration for headless sites).
- **GBP automation:** posts, Q&A responses, attribute updates, review reply drafts.
- **Citation-earning playbooks:** auto-generated digital PR target lists (which domains AI engines cite in your category), outreach drafts, and third-party profile completeness tasks (G2, Capterra, JustDial, etc. by market).

Every action is: proposed → previewed (diff view) → approved (one click or auto-approve rules) → deployed → measured (did the fix move the metric?). This closes the loop no competitor closes.

Pillar D — Intelligence & Reporting Layer

- **AI Copilot:** natural-language querying of all platform data ("Why did our AI citations drop in Germany last month?") with cited, drill-downable answers.
- **Attribution & forecasting:** AI-referral revenue attribution (GA4 + server-side tagging), ranking→traffic→revenue forecasting for executive buy-in, anomaly detection tied to algorithm-update timelines.
- **Reporting:** white-label PDF/live dashboards, Looker Studio connector, scheduled client reports, agency multi-workspace management.
- **API + MCP server:** full data API, plus an MCP server so customers' own AI agents (Claude, ChatGPT) can query and act on their SEO data — makes the platform "agent-native" from day one.

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3. System Architecture

Four layers, event-driven, built to scale from 100 customers to 100,000.

Layer 1 — Data Acquisition

SERP data: Managed APIs first (DataForSEO ~$0.60/1k queries as primary; SerpAPI as failover). Do NOT build a Google scraper in year one — parser maintenance and legal exposure aren't worth it. Revisit at >$50k/month API spend.

LLM answer polling: Adapter-per-engine architecture behind a common interface:

- API-based: OpenAI, Gemini, Perplexity (Sonar), Anthropic — cheap, but API answers differ from consumer-app answers.

- Consumer-surface capture: headless browser sessions against consumer interfaces for ground truth on a sampled basis (higher cost, higher fidelity). Run APIs daily, consumer capture weekly on a stratified sample; statistically reconcile the two.
- **Non-determinism handling:** every prompt runs n=3–5 times per cycle; report citation rates as ranges with confidence bands, never point estimates. This honesty is itself a differentiator.

Crawling: Distributed cloud crawler (Playwright cluster on Kubernetes) with JS rendering, polite rate-limiting, per-project crawl budgets. Start capped at 100k URLs/project; enterprise tier lifts caps.

First-party connectors: Google Search Console (including the new Generative AI performance reports), GA4, Bing Webmaster, GBP API, CMS plugins (WordPress, Shopify, Webflow), GitHub.

Backlink/mention data: license initially (DataForSEO backlinks or similar); build a web-scale mention index (not link index) in v2 — mentions matter more for AI visibility and are cheaper to index than full link graphs.

Layer 2 — Processing & Intelligence

- **Stream ingestion:** Kafka (or Redpanda) topics per data type → stream processors normalize into the warehouse.
- **NLP pipeline:** entity extraction (spaCy + fine-tuned multilingual transformer; IndicBERT/MuRIL family for Indian languages), embedding generation (multilingual-e5 class models, self-hosted for cost), intent classification, sentiment on AI answers, keyword clustering (HDBSCAN on embeddings).
- **Scoring models:** technical health score, extractability score (trained on pages that do vs don't get cited — this proprietary training signal compounds over time into a real data moat), entity strength score, unified visibility score.
- **LLM orchestration:** one internal gateway service routing to the cheapest capable model per task (small model for classification, frontier model for content generation and copilot), with aggressive caching (identical prompt-bank runs cached across customers in the same category — massive cost lever).

Layer 3 — Storage (the data layering)

Store	Technology	Holds
Operational	PostgreSQL	Accounts, projects, configs, fix-queue state, billing
Analytical/time-series	ClickHouse	SERP positions, AI citation events, crawl results, metrics history — petabyte-capable, sub-second aggregations for dashboards
Object/raw lake	S3 + Parquet	Raw SERP snapshots + raw AI answers, 24-month retention (enables reprocessing when scoring models

Store	Technology	Holds
		improve — retroactive intelligence)
Vector	pgvector → dedicated (Qdrant) at scale	Embeddings for content, keywords, entities, AI answers
Cache/queue	Redis	Sessions, rate limits, job queues (BullMQ/Celery)

Entity-first data model: the core join key is not the URL — it's the entity. `Entity → {URLs, keywords, prompts, citations, mentions, schema, Wikidata ID}`. This single design decision makes the SEO and GEO worlds queryable as one, and is what most incumbents can't retrofit.

Layer 4 — Presentation & Action

- Web app (see §5), REST + GraphQL API, MCP server, webhook/alert system (Slack, email, Teams), CMS plugins and Cloudflare Worker for edge deployment of fixes.

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4. AI Components Summary

Component	Approach	Cost control
AI visibility polling	Multi-engine adapters, n-sampling	Category-level prompt caching across customers
Copilot	Frontier LLM + RAG over customer's own data in ClickHouse	Semantic caching, small-model routing
Content generation	Frontier LLM grounded in entity/SERP data, brand-voice fine-tune per enterprise customer	Draft with mid-tier model, polish with frontier
Extractability scorer	Proprietary fine-tuned classifier trained on cited vs non-cited pages	Self-hosted inference
Multilingual NLP	MuRIL/IndicBERT-class for Indic; multilingual-e5 embeddings	Self-hosted GPU pool
Anomaly detection	Statistical (Prophet-class) on ClickHouse metrics	Cheap, no LLM

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5. Frontend & UX System ("best looking, easy to use, highly efficient")

Design philosophy: "One number, then depth." Every screen leads with a single comprehensible answer, with progressive disclosure into full data. This directly attacks the incumbents' two biggest UX failures: overwhelming dashboards and 12-click navigation.

Information architecture (max 2 levels deep):

1. **Home / Pulse** — the Unified Visibility Score, its 30-day trend, top 3 wins, top 3 risks, and the Fix Queue count. A business owner understands this screen in 10 seconds.
2. **Visibility** — organic rankings, AI citations, local — all in one section with a shared filter bar (market, language, engine, device).
3. **Fix Queue** — the product's heart. Kanban-style: Proposed → Approved → Deployed → Verified, each card showing predicted impact and effort. This is where users live.
4. **Content Studio** — briefs, drafting, extractability scoring, decay alerts.
5. **Reports & Copilot** — persistent copilot available on every screen (⌘K), not buried in a tab.

Design system specifics:

- Stack: React + TypeScript, Tailwind + a custom token system, Radix primitives, TanStack Query/Table, ECharts or visx for visualization (fast canvas rendering for large time-series), Framer Motion used sparingly.
- Visual language: light-first with true dark mode; one accent color; data-ink-maximized charts (no gradients-for-decoration); confidence bands rendered on all AI-visibility charts (honesty as a visual signature).
- **Role-adaptive views:** the same data renders as "Owner mode" (plain-language, prioritized actions), "Analyst mode" (full tables, exports, query builder), "Executive mode" (trend + revenue attribution), "Agency mode" (multi-client grid, white-label toggle). Set at user level, switchable instantly.
- Performance budget: first meaningful paint <1.5s, every dashboard query <500ms (ClickHouse makes this achievable), skeleton loading everywhere, offline-tolerant report viewing.
- Onboarding: connect domain + GSC → first insights in <10 minutes → first proposed fix in <48 hours. Time-to-first-value is a tracked product KPI.
- Full i18n from day one (UI in English, Hindi, Spanish, Portuguese, Japanese, German at launch; RTL-ready).

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6. Pricing & Go-to-Market

Transparent, non-credit pricing (a direct strike at Ahrefs' credit backlash):

Tier	Global price	India/emerging price	Includes
Starter	$49/mo	₹1,999/mo	1 project, 500 keywords, 100 AI prompts, weekly polling, Fix Queue (manual apply)
Growth	$149/mo	₹5,999/mo	5 projects, 2,500 keywords, 500 prompts, daily polling, auto-deploy integrations, local SEO
Agency	$399/mo	₹14,999/mo	25 projects, white-label, client workspaces, unlimited seats
Enterprise	Custom ($20-60k/yr)	Custom	Log analysis, SSO, API/MCP, unlimited scale, brand-voice fine-tune, SLA

Regional pricing is enforced via billing address + payment method (UPI/Razorpay for India, Stripe globally). Free tier: one project, 25 keywords, 10 prompts, weekly — a genuine trial, and a data-acquisition engine for the scoring models.

GTM sequencing: (1) launch into the GEO-curious SEO community with a free "AI Visibility Report" lead magnet (enter domain → instant citation snapshot); (2) agencies as the multiplier channel (white-label + margin); (3) India SMB via partnerships (Razorpay/Zoho ecosystems, vernacular content marketing); (4) enterprise via the attribution/forecasting story once case studies exist.

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7. Feasibility, Costs & Risks

Unit economics at 1,000 paying customers (rough):

- SERP API: ~$8–15k/mo · LLM polling: ~$10–20k/mo (before caching; caching cuts 40-60%) · Infra (K8s, ClickHouse, GPU pool): ~$8–12k/mo · Total COGS ≈ $30–45k/mo against ~$120–150k MRR at blended $130 ARPU → 65–75% gross margin. Viable.

Top risks and mitigations:

Risk	Mitigation
Google restricts SERP data access further	Multi-vendor API abstraction; shift weight to first-party (GSC) data; the AI-visibility side doesn't depend on Google SERPs
LLM providers block programmatic answer capture	API-first polling (sanctioned); consumer capture only sampled; partnerships where offered (Perplexity has a publisher/API posture)
AI answer non-determinism undermines trust	Confidence bands, n-sampling, "directional not exact" framing baked into UI
Incumbent bundling (Adobe-Semrush)	Speed + price + execution layer; incumbents can't retrofit entity-first data models or agentic deployment quickly
Execution layer breaks a customer's site	Human-in-the-loop default, staged rollouts, automatic rollback, edge-worker sandboxing, insurance-grade audit logs
India ARPU too low to serve profitably	Automation-first support, self-serve onboarding, shared caching economics, UPI billing

Legal checklist: ToS review for every data vendor and LLM provider; GDPR + DPDP + CCPA compliance (data residency options: EU + India regions on the roadmap); clear customer data ownership terms; no training on customer private data without opt-in.

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8. Roadmap & Team

Phase 1 — MVP (months 0–6), team of 6–8: Unified visibility (rank tracking + AI polling across 5 engines), technical audit, Fix Queue with schema + meta deployment via WordPress/Shopify plugins, Pulse dashboard, Starter/Growth tiers, English + Hindi keyword support. *Exit criteria: 50 design partners, demonstrable citation-rate lift for ⩾10 of them, <48h time-to-first-fix.*

Phase 2 — Depth (months 6–18), team of 12–18: Content Studio + extractability scorer, entity/knowledge-graph engine, backlink/mention licensing, local SEO suite, agency white-label, Copilot, 6 UI languages, India pricing launch, GSC generative-report ingestion, GitHub PR deployment. *Exit criteria: $1.5M ARR, agency logos, published case studies with revenue attribution.*

Phase 3 — Platform (months 18–36), team of 25+: Log-file analysis + enterprise tier, ACP/UCP agent-ready commerce feeds, MCP server GA, proprietary mention index, forecasting suite, Baidu/Naver/Yandex tracking, EU + India data residency. *Exit criteria: first $50k+ enterprise contracts, measurable agent-referred revenue for e-commerce customers.*

Kill/pivot triggers: if Google ships clean AI-click data in GSC, drop proprietary AI-CTR estimation and double down on execution; if one answer engine takes >50% of AI referrals, concentrate polling there; if SERP API costs spike >3x, accelerate first-party-data strategy.

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9. What Makes This Win (summary)

1. **Entity-first data model** — the architectural decision incumbents can't retrofit.
2. **The Fix Queue** — the only platform that deploys, verifies, and rolls back fixes, not just reports problems.
3. **Honest AI metrics** — confidence bands and n-sampling build trust in a category full of false precision.
4. **One price, everything visible** — SEO + GEO + local unified, no credits, region-fair pricing.
5. **Vernacular + multi-engine** — structurally underserved markets (India first) as the growth wedge.
6. **Agent-native** — MCP server and ACP/UCP tooling position it for the agentic-commerce era rather than just the current one.